

```

class Node:
    def __init__(self, data, level, fval):
        """ Initialize the node with the data, level of the node and the calculated fvalue """
        self.data = data
        self.level = level
        self.fval = fval

    def generate_child(self):
        """ Generate child nodes from the given node by moving the blank space
            either in the four directions {up,down,left,right} """
        x, y = self.find(self.data, '_')
        """ val_list contains position values for moving the blank space in either of
            the 4 directions [up,down,left,right] respectively. """
        val_list = [[x, y-1], [x, y+1], [x-1, y], [x+1, y]]
        children = []
        for i in val_list:
            child = self.shuffle(self.data, x, y, i[0], i[1])
            if child is not None:
                child_node = Node(child, self.level+1, 0)
                children.append(child_node)
        return children

    def shuffle(self, puz, x1, y1, x2, y2):
        """ Move the blank space in the given direction and if the position value are out
            of limits the return None """
        if x2 >= 0 and x2 < len(self.data) and y2 >= 0 and y2 < len(self.data):
            temp_puz = []
            temp_puz = self.copy(puz)
            temp = temp_puz[x2][y2]
            temp_puz[x2][y2] = temp_puz[x1][y1]
            temp_puz[x1][y1] = temp
            return temp_puz
        else:
            return None

    def copy(self, root):
        """ Copy function to create a similar matrix of the given node"""
        temp = []
        for i in root:
            t = []
            for j in i:
                t.append(j)
            temp.append(t)
        return temp

    def find(self, puz, x):
        """ Specifically used to find the position of the blank space """
        for i in range(0, len(self.data)):
            for j in range(0, len(self.data)):
                if puz[i][j] == x:
                    return i, j

class Puzzle:
    def __init__(self, size):
        """ Initialize the puzzle size by the specified size, open and closed lists to empty """
        self.n = size
        self.open = []
        self.closed = []

    def accept(self):
        """ Accepts the puzzle from the user """
        puz = []
        for i in range(0, self.n):
            temp = input().split(" ")
            puz.append(temp)
        return puz

    def f(self, start, goal):
        """ Heuristic Function to calculate hueristic value f(x) = h(x) + g(x) """
        return self.h(start.data, goal) + start.level

    def h(self, start, goal):
        """ Calculates the different between the given puzzles """
        temp = 0
        for i in range(0, self.n):
            for j in range(0, self.n):
                if start[i][j] != goal[i][j] and start[i][j] != '_':
                    temp += 1
        return temp

    def process(self):

```

```

def process(self):
    """ Accept Start and Goal Puzzle state"""
    print("Enter the start state matrix \n")
    start = self.accept()
    print("Enter the goal state matrix \n")
    goal = self.accept()

    start_node = Node(start, 0, 0)
    start_node.fval = self.f(start_node, goal)
    """ Put the start node in the open list"""
    self.open.append(start_node)
    while True:
        cur = self.open[0]
        print("")
        print(" | ")
        print(" \\'/ \n")
        for i in cur.data:
            for j in i:
                print(j, end=" ")
            print("")
        """ If the difference between current and goal node is 0 we have reached the goal node"""
        if self.h(cur.data, goal) == 0:
            print("Goal state reached successfully")
            break
        for i in cur.generate_child():
            i.fval = self.f(i, goal)
            self.open.append(i)
        self.closed.append(cur)
        del self.open[0]
        """ sort the open list based on f value """
        self.open.sort(key=lambda x: x.fval, reverse=False)

puz = Puzzle(3)
puz.process()

```

```

2 8 3
1 6 4
7 _ 5
Enter the goal state matrix

```

```

1 2 3
8 _ 4
7 6 5

```

```

|
\ '/'

```

```

2 8 3
1 6 4
7 _ 5

```

```

|
\ '/'

```

```

2 8 3
1 _ 4
7 6 5

```

```

|
\ '/'

```

```

2 8 3
_ 1 4
7 6 5

```

```

|
\ '/'

```

```

2 _ 3
1 8 4
7 6 5

```

```

|
\ '/'

```

```

_ 2 3
1 8 4
7 6 5

```

```

|
\ '/'

```

```

1 2 3
_ 8 4
7 6 5

```

```
\ /  
1 2 3  
8 _ 4  
7 6 5  
Goal state reached successfully
```